

Quiz2 - Results



Attempt 1 of 1

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Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

Which of the following is true about using CNNs and MLPs for text classification?

- ☐ Only MLP's can handle polysemy.
- ☒ Only CNN's are sensitive to word order.
- ☐ Only MLP's require using a mean pooling layer.
- ☐ Only CNN's use a non-linear activation function.

Question 2

1 / 1 point

For multi-class classification, what changes compared to the 2-layer neural network for binary classification?

- ☐ Both hidden and final layers must be replaced.
- ☐ The activation function must be removed.

- ☐ Only the hidden layer changes.
- ✓ ☒ Only the final layer changes (to softmax regression).

Question 3

1 / 1 point

Another limitation mentioned for classic word vectors is difficulty with:

- ☐ Building vocabulary lists.
- ☐ Any arithmetic operations.
- ☐ Using gradient descent.
- ✓ ☒ Polysemy (a word having multiple meanings).

Question 4

1 / 1 point

When sampling negative examples in word2vec, why is the α -weighted frequency preferred over raw frequency?

- ☐ To ensure that extremely rare words are never sampled as negatives.
- ☐ To increase the dimensionality of the word vectors during the training process.
- ☐ To ensure that the gradient updates for context vectors are always positive.
- ✓ ☒ To down-weight extremely common words so they aren't over-represented in the negative samples.

Question 5

1 / 1 point

Why is logistic regression optimization convex, but word2vec optimization not convex?

- ☐ Logistic regression uses cosine similarity; word2vec uses dot products.
- ☐ Logistic regression has no gradients; word2vec does.
- ☐ Logistic regression requires negative sampling; word2vec does not.
- ✓ ☒ Logistic regression optimizes only weights with fixed features; word2vec optimizes both "features" (word vectors) and "weights" (context vectors).

Question 6

1 / 1 point

Mean pooling for text classification means:

- ☐ Taking the max over each embedding dimension only on nouns.
- ☐ Concatenating all word vectors into one long vector.
- ☐ Sorting word vectors and taking the median word.
- ✓ ☒ Averaging the word vectors in the text to get a fixed-length representation.

Question 7

1 / 1 point

Why is inserting activation functions (e.g., sigmoid) to the neural network important?

- ☐ It removes the need for gradient descent.
- ✓ ☒ Without them, the whole network is a linear function.

- ☐ It ensures that training converges to a global minimum.
- ☐ It helps the model process documents with variable numbers of words.

Question 8

1 / 1 point

The popular optimizer 'Adam' is essentially a combination of which two concepts?

- ✓ ☒ Momentum and RMSProp
- ☐ L1 and L2 regularization
- ☐ Sigmoid activations and Linear layers
- ☐ Backpropagation and the Chain Rule

Question 9

1 / 1 point

What is a major risk of setting the learning rate too high during neural network training?

- ☐ The model will automatically revert to a linear classifier.
- ☐ The model is more likely to become deeper during training.
- ☐ It will cause the weight matrices to become sparse.
- ✓ ☒ It will cause drastic updates which lead to divergent behaviors.

Question 10

1 / 1 point

Compute cosine similarity between $u=[1,2]$ and $v=[2,0]$.

 ≈ 0.707 

0

 ≈ 0.447  ≈ 0.894 **Done**